

Current Cybersecurity, Cryptography & Cyber Law

Lesson 64 — Crypto, Firewalls and Law

JKSSB Computer Awareness

October 8, 2026

Where This Lesson Fits

- ▶ The syllabus also tests **computer virus & latest anti-virus, terms and abbreviations used in IT**, and the **role of IT in governance**.
- ▶ This lesson covers the current-affairs edge of that scope: threats, crypto basics, firewalls, and cyber law at **awareness level**.
- ▶ Depth stays at recall and classification (L1/L2), with one short scenario at most.

The one idea

Know what each threat, tool, and term **does** — and what its nearest neighbour does instead.

Malware Family: Virus First

- ▶ **Malware** is hostile software: virus, worm, Trojan horse, ransomware, spyware.
- ▶ A **virus** needs a host program and human action to spread; a **worm** spreads on its own across a network.
- ▶ A **Trojan horse** looks like useful software but hides a harmful payload; it does **not** self-replicate.

Keep the pair straight

Virus needs a host and help to spread; worm spreads by itself.

Parasitic (File-Infecter) Virus

- ▶ A **parasitic virus**, also called a **file-infecter virus**, attaches itself to an executable file.
- ▶ When the infected program runs, the virus code runs **first**, then hands control back to the host program.
- ▶ It spreads as infected files are **shared or copied** to other machines.
- ▶ An **anti-virus** detects, blocks, and removes such infections; its database must be kept **updated**.

Trap alert

A file infecter attaches to files. It is not a worm and not a Trojan.

Four Threats Side by Side

Threat	What happens
Phishing	Fraudulent mail or message tricks the user into revealing passwords or OTPs
Ransomware	Malicious code locks or encrypts files, then demands payment
DDoS	Flood of traffic from many machines overwhelms a server
Hacking	Unauthorised access to a computer or account

Each row is a different mechanism; the exam swaps their definitions.

Phishing vs Ransomware vs DDoS vs Hacking

- ▶ **Phishing** attacks the **user**: fake bank or KYC mail, fake login page, stolen credentials.
- ▶ **Ransomware** attacks the **data**: files are encrypted and a ransom is demanded for the key.
- ▶ **DDoS** (Distributed Denial of Service) attacks **availability**: the service is flooded until genuine users cannot reach it.
- ▶ **Hacking** is the broad term for **unauthorised access**; the other three are specific methods or goals.

Office desktop example

A clerk gets a fake “verify your account” mail (phishing), while the office server slows to a halt under flood traffic (DDoS) — different threats, different defences.

Cryptography in One Frame

- ▶ **Cryptography** protects information by transforming it so only the intended recipient can read it.
- ▶ **Plaintext** is the readable message; **ciphertext** is the transformed message; a **key** controls the transformation.
- ▶ **Encryption** converts plaintext to ciphertext; **decryption** converts it back.
- ▶ Two broad families: **symmetric** (same key both sides) and **asymmetric** (public key and private key pair).

DES: The 64-Bit Block Fact

- ▶ **DES** stands for Data Encryption Standard.
- ▶ It is a **symmetric** cipher: the same key encrypts and decrypts.
- ▶ DES works on **64-bit blocks** of data at a time.
- ▶ Its key is 64 bits long, of which **56 bits** are effective (8 bits are parity).

Trap alert

DES block size is **64 bits**. Do not confuse block size with the 56-bit effective key.

Encryption vs Digital Signature (TRAP-15)

- ▶ **Encryption** provides **confidentiality**: only the intended recipient can read the message.
- ▶ A **digital signature** provides **authenticity and integrity**: it proves who sent the message and that it was not altered.
- ▶ A signature does **not** hide the message content; encryption does **not** by itself prove the sender's identity.

Trap alert

Digital signature $\neq$ encryption. Signature = who sent it and unchanged; encryption = who can read it. (TRAP-15/TRM-11)

Firewalls: What They Do

- ▶ A **firewall** is a barrier between a trusted internal network and an untrusted external network.
- ▶ It permits or denies traffic according to a **rule set**.
- ▶ It filters **packets and connections**, not file-infecting viruses inside allowed downloads — that is the anti-virus's job.
- ▶ Three types tested: packet-filtering, stateful inspection, and application (proxy) firewalls.

Three Firewall Types

Type	Checks	Level
Packet-filtering	Source/destination address, port, protocol	Network layer, per packet
Stateful inspection	Same headers plus connection state	Tracks sessions, not single packets
Application / proxy	Content and commands of the application	Application layer; acts as go-between

Order of depth: packet header only, then header plus state, then full application content.

Firewall Ordering Drill

- ▶ **Packet-filtering**: fastest and simplest; decides packet by packet; cannot judge a session as a whole.
- ▶ **Stateful inspection**: remembers open connections; drops packets that do not belong to a known session.
- ▶ **Application/proxy firewall**: sits between client and server and inspects application commands (for example web or mail requests).
- ▶ Exam shorthand: header only → header plus state → full content via proxy.

Keep the triple straight

Filtering checks the envelope; stateful checks the conversation; proxy reads the letter inside.

Internet Governance Concerns

- ▶ The Internet has **no single owner**; addressing and names are coordinated internationally (for example by ICANN for names and numbers).
- ▶ Governance concerns tested at awareness level: **security** of networks, **privacy** of personal data, fair **access**, and which country's **law** applies to cross-border acts.
- ▶ The **role of IT in governance** (e-governance) is the positive side: online services, transparency, and delivery to citizens.

Privacy and Data Protection

- ▶ **Personal data** is any data that identifies a person: name with phone number, biometrics, account details.
- ▶ Protection principles at awareness level: collect with **consent**, use only for the stated **purpose**, keep data **secure**, and retain it no longer than needed.
- ▶ **Privacy settings**, strong passwords, and OTP discipline are the user-side defences the exam names.
- ▶ Leaking or misusing personal data can also attract **legal** consequences under cyber law.

Copyright Act: Awareness-Level Map

- ▶ The **Copyright Act** protects original literary, artistic, musical, and software works from copying without permission.
- ▶ Awareness-level mapping only: **meaning of copyright** (what rights the owner holds), **first ownership** (who owns it first), **infringement** (what counts as violation), **fair dealing** (limited permitted uses such as study), **offences** (punishment for wilful infringement).
- ▶ Software piracy — copying licensed software without permission — is the exam's standard infringement example.

Exam depth

Match the **concept** to its section area; section numbers are recall only where the question itself supplies them.

Abbreviation and Term Ladder

- ▶ **DES** — Data Encryption Standard; 64-bit blocks.
- ▶ **DDoS** — Distributed Denial of Service; flood from many machines.
- ▶ **OTP** — One Time Password; phishing target.
- ▶ **ICANN** — Internet Corporation for Assigned Names and Numbers.
- ▶ **IP** — Internet Protocol; the address a packet filter reads.

Trap alert

DDoS needs many machines (distributed); a single-machine flood is only DoS.

Trap Alert: Facts to Keep Straight

- ▶ DES block size is **64 bits** (56-bit effective key) — not 128.
- ▶ Digital signature $\neq$ encryption: authenticity/integrity vs confidentiality. (TRAP-15)
- ▶ File infector attaches to **executable files**; it is not a worm.
- ▶ Packet filter checks headers only; stateful adds **connection state**; proxy inspects **application content**.
- ▶ Phishing steals **credentials**; ransomware locks **files**; DDoS blocks **access**; hacking is unauthorised access generally.

Lesson 64: Recall

- ▶ A parasitic (file-infector) virus attaches to executable files and runs when the host program runs.
- ▶ DES (Data Encryption Standard) is symmetric and works on 64-bit blocks.
- ▶ Firewalls: packet-filtering (headers), stateful (plus connection state), application/proxy (application content).
- ▶ Phishing tricks the user; ransomware encrypts files for ransom; DDoS floods a service; hacking is unauthorised access.
- ▶ Encryption gives confidentiality; a digital signature gives authenticity and integrity.
- ▶ Governance concerns: security, privacy, access, jurisdiction; copyright protects owners against infringement, with fair-dealing exceptions.